Object Oriented Programming

Inheritance

Lab – 05

Lab Tasks:

Exercise 1.1: Create a class Employee with the following attributes and member functions:

- Employee registration number
- Employee name
- Designation
- Human resource Allowance
- Basic salary
- Profitable fund
- Get function to take input of Registration number, name and designation.

Create another class salary that inherits using **protected inheritance** from employee. Salary should contain the following member functions:

- Get function to take input of human resource allowance, basic salary and profitable fund.

Exercise 1.2: Write a Class person that has attributes of id, name and address. It has constructor to initialize and member function to input and member function to display data members. Create second class student that inherit from person class. It has additional attributes of roll number and marks. It has constructor to initialize, member function to input and display data members. Create third class Scholarship that inherit from student class. It has additional attributes of scholarship name and amount. It also has member function input and display.

Exercise 1.3: Create two classes named Mammals and MarineAnimals. Create another class named BlueWhale which inherits both the above classes. Now, create a function in each of these classes which prints "I am mammal", "I am a marine animal" and "I belong to both the categories: Mammals as well as Marine Animals" respectively. Now, create an object for each of the above class and try calling

1 - function of Mammals by the object of Mammal

2 - function of MarineAnimal by the object of MarineAnimal

3 - function of BlueWhale by the object of BlueWhale

4 - function of each of its parent by the object of BlueWhale.

Exercise 1.4: Imagine a publishing company that markets both book and audiocassette versions of its works. Create a class publication that stores the title (a string) and price (type float of a publication. From this class derive two classes: **book,** which adds a page count (type int), and **tape,** which adds a playing time in minutes (type float). Each of these three classes should have a getdata() function to get its data from the user at the keyboard, and a putdata() function to display its data. Write a main() program to test the book and tape classes by creating instances of them, asking the user to fill in data with getdata(), and then displaying the data with putdata().

Exercise 1.5: Start with the publication, book, and tape classes of Exercise 1. Add a base class sales that holds an array of three floats so that it can record the dollar sales of a particular publication for the last three months. Include a getdata() function to get three sales amounts from the user, and a putdata() function to display the sales figures. Alter the book and tape classes so they are derived from both publication and sales. An object of class book or tape should input and output sales data along with its other data. Write a main() function to create a book object and a tape object and exercise their input/output capabilities.

Exercise 1.6: Make a class named Fruit with a data member to calculate the number of fruits in a basket. Create two other class named Apples and Mangoes to calculate the number of apples and mangoes in the basket. Print the number of fruits of each type and the total number of fruits in the basket.

We want to store the information of different vehicles. Create a class named Vehicle with two data member named mileage and price. Create its two subclasses

*Car with data members to store ownership cost, warranty (by years), seating capacity and fuel type (diesel or petrol).

*Bike with data members to store the number of cylinders, number of gears, cooling type(air, liquid or oil), wheel type(alloys or spokes) and fuel tank size(in inches)

Make another two subclasses Audi and Ford of Car, each having a data member to store the model type. Next, make two subclasses Bajaj and TVS, each having a data member to store the make-type.

Now, store and print the information of an Audi and a Ford car (i.e. model type, ownership cost, warranty, seating capacity, fuel type, mileage and price.) Do the same for a Bajaj and a TVS bike.

Note: Do Practise of programming exercise given in book chapter 9 of Object Oriented Programming in C++ by Robert Lafore.